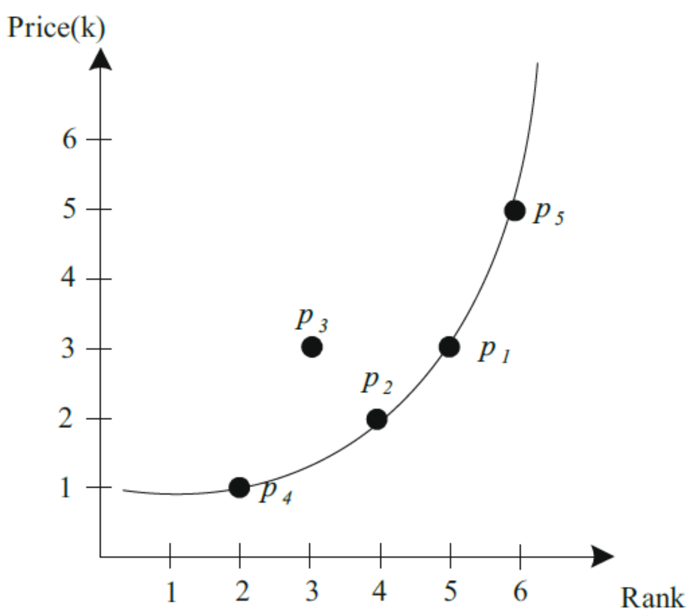
Skyline query

- Finding points not dominated by any other object, in terms of *m* dimensions
  - *p* dominates *p*' if its value in each dimension is as good or better than that of *p*' and is better in at least one dimension



hotel	rank	price(k)
p_1	5	3
p_2	4	2
p_3	3	3
p_4	2	1
p_5	6	5

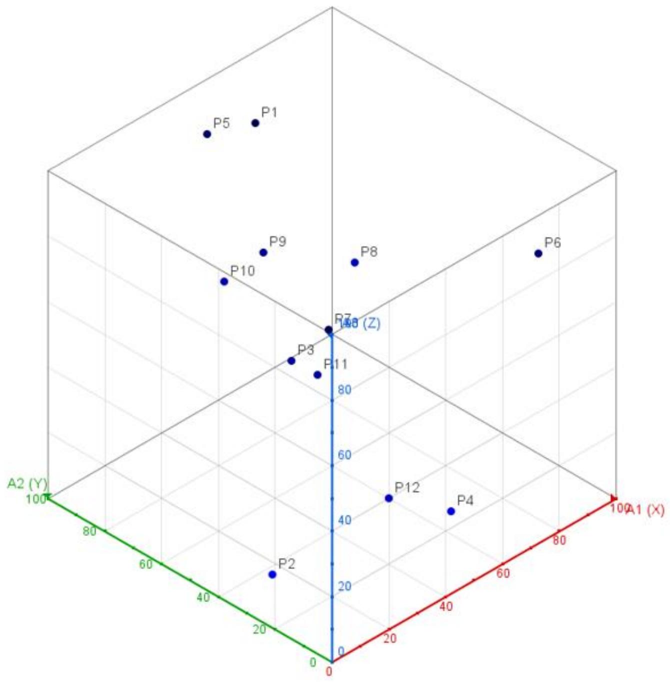
p_1 dominates p_3

輸入資料

- Input: point.txt

值介於1~100

id	A1	A2	A3
P1	52	79	99
P2	11	32	5
P3	28	42	57
P4	52	10	15
P5	53	97	86
P6	85	12	76
P7	6	7	95
P8	90	82	36
P9	48	72	65
P10	58	96	39
P11	30	35	55
P12	31	11	29



建構 3D R-tree

```
Level 0: Node MBR=(6,7,5)-(90,97,99) size=2
  Level 1: Node MBR=(52,10,15)-(53,97,99) size=1
    Level 2: Node MBR=(52,10,15)-(53,97,99) size=1
      Level 3: Node MBR=(52,10,15)-(53,97,99) size=2
        Level 4: Node MBR=(52,10,15)-(52,79,99) size=2
          Level 5: P4 52 10 15
          Level 5: P1 52 79 99
        Level 4: Node MBR=(53,96,39)-(58,97,86) size=2
          Level 5: P5 53 97 86
          Level 5: P10 58 96 39
  Level 1: Node MBR=(6,7,5)-(90,82,95) size=2
    Level 2: Node MBR=(6,7,76)-(85,12,95) size=1
      Level 3: Node MBR=(6,7,29)-(85,12,95) size=2
        Level 4: Node MBR=(6,7,95)-(6,7,95) size=1
          Level 5: P7 6 7 95
        Level 4: Node MBR=(31,11,29)-(85,12,76) size=2
          Level 5: P6 85 12 76
          Level 5: P12 31 11 29
    Level 2: Node MBR=(11,32,5)-(90,82,57) size=2
      Level 3: Node MBR=(90,82,36)-(90,82,36) size=1
        Level 4: Node MBR=(48,72,36)-(90,82,65) size=2
          Level 5: P8 90 82 36
          Level 5: P9 48 72 65
      Level 3: Node MBR=(11,32,5)-(30,42,57) size=2
        Level 4: Node MBR=(11,32,5)-(11,32,5) size=1
          Level 5: P2 11 32 5
        Level 4: Node MBR=(28,35,55)-(30,42,57) size=2
          Level 5: P3 28 42 57
          Level 5: P11 30 35 55
```

執行 skyline query

id	A1	A2	A3
P1	52	79	99
P2	11	32	5
P3	28	42	57
P4	52	10	15
P5	53	97	86
P6	85	12	76
P7	6	7	95
P8	90	82	36
P9	48	72	65
P10	58	96	39
P11	30	35	55
P12	31	11	29